

WARTEQ - Machine Learning–Based Spatial Clustering for Lake Water Quality Monitoring

Clayvent Yuuki William
Binus ASO School of Engineering
Tangerang, Alam Sutera

Dwi Naya Syah Hary
Binus ASO School of Engineering
Tangerang, Alam Sutera

Louis Oliver
Binus ASO School of Engineering
Tangerang, Alam Sutera

Rasyhad Aqil Wiryawan
Binus ASO School of Engineering
Tangerang, Alam Sutera

Avriel Zara Xaviera
Binus ASO School of Engineering
Tangerang, Alam Sutera

Dr. Lukas, ST, MAI, CISA
Binus ASO School of Engineering
Tangerang, Alam Sutera

ABSTRACT

Conventional water-quality monitoring often relies on manual sampling at limited points, which may fail to capture spatial heterogeneity across large water bodies. This paper presents the machine learning component of **WARTEQ**, an RC-boat-based lake monitoring system that applies unsupervised clustering to in-situ sensor measurements. A six-feature representation is constructed from pH, temperature, turbidity, and total dissolved solids (TDS), including logarithmic transformations of turbidity and TDS to reduce skewness. Features are standardized using Z-score normalization and clustered using the K-Means algorithm. The number of clusters is selected using the Elbow Method and validated using internal cluster validity metrics, including the Silhouette coefficient, Davies–Bouldin Index, and Calinski–Harabasz Index. The trained model is deployed in a browser-based dashboard by exporting feature-scaling parameters and cluster centroids for real-time inference. Experimental results demonstrate a consistent three-group separation that supports an interpretable, relative ranking of water-quality conditions (Good, Moderate, Poor) within a given lake survey.

Index Terms— K-Means, clustering, water quality, IoT, environmental monitoring, PCA, spatial heterogeneity.

1. INTRODUCTION

Freshwater quality is essential for public health, biodiversity, and economic activity. However, increasing

pollution from domestic discharge, agricultural runoff, and industrial effluents has significantly degraded water resources worldwide [1]. Lakes, in particular, exhibit strong spatial heterogeneity due to inflow points, sediment transport, and localized anthropogenic activity. As a result, point-based sampling strategies may fail to represent true spatial conditions across a water body.

WARTEQ addresses this limitation by collecting geotagged water-quality measurements using an RC boat and summarizing spatial patterns through a web-based dashboard. Instead of relying on supervised classification—which requires labeled ground-truth data obtained through laboratory analysis—WARTEQ employs unsupervised learning to discover structure directly from sensor measurements. Clustering is well suited for this task when the objective is to identify natural regimes and relative differences within a survey rather than enforce predefined regulatory classes [2].

2. OBJECTIVES

This study has four main objectives:

1. to construct a robust feature representation from on-board sensor measurements (pH, temperature, turbidity, and TDS);
2. to determine an appropriate number of clusters that reflect spatial regimes;
3. to evaluate clustering quality using established internal validation metrics; and

4. to deploy the trained model into a lightweight, browser-based inference pipeline suitable for real-time visualization.

3. DATASET AND VARIABLES

The clustering model is trained using an open-source water-quality dataset published in CSV format on GitHub [10]. The dataset contains measurements of pH, temperature (°C), turbidity (NTU), and either total dissolved solids (TDS, ppm) or electrical conductivity (µS/cm). Records with missing values in any required field are removed to ensure consistent feature vectors. WARTEQ uses the same data schema when ingesting live measurements from Firebase.

When TDS measurements are unavailable but conductivity values are provided, TDS is estimated using a widely used freshwater approximation:

$$\text{TDS} \approx 0.64 \times \text{Conductivity}$$

where EC is electrical conductivity in µS/cm. This empirical factor is commonly applied in freshwater monitoring and provides a reasonable approximation depending on ionic composition [7].

The raw water-quality measurements exhibit substantial variation across samples, particularly for turbidity and total dissolved solids (TDS), which are commonly characterized by right-skewed distributions due to the presence of high-solid measurements. Such skewness motivates the application of feature transformations to improve the stability of distance-based clustering algorithms.

4. MACHINE LEARNING METHODOLOGY

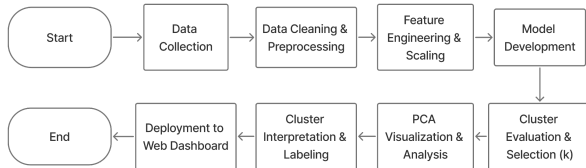


Fig.1 Step-by-step Methodology Flowchart

A. Data Cleaning and Preprocessing

Each water-quality measurement is initially represented as a vector:

$$\mathbf{x} = [\text{pH}, \text{Temp}, \text{Turb}, \text{TDS}]$$

The dataset is validated to ensure all required columns exist. Rows containing missing values are removed, and all features are converted to floating-point format to prevent type inconsistencies during scaling and clustering. This process mirrors the implemented training script, which logs dataset dimensions before and after data cleaning to ensure traceability.

B. Feature Engineering

Environmental variables such as turbidity and TDS often follow non-Gaussian distributions with heavy right tails. Because K-Means relies on Euclidean distance, extreme values can disproportionately influence centroid assignment [3]. To mitigate this effect, logarithmic transformations are applied.

$$\begin{aligned} \log_Turb &= \log(1 + \text{Turb}) \\ \log_TDS &= \log(1 + \text{TDS}) \end{aligned}$$

The final feature vector used for clustering is therefore:

$$\mathbf{f} = [\text{pH}, \text{Temp}, \text{Turb}, \text{TDS}, \log(1+\text{Turb}), \log(1+\text{TDS})]$$

Both raw and logarithmically transformed versions of turbidity and TDS are retained in the feature vector. The raw features preserve absolute magnitude differences relevant to pollution load, while the log-transformed features reduce skewness and emphasize relative variation among low-to-moderate values. This multi-scale representation allows K-Means to account for both absolute and relative differences in water-quality characteristics.

C. Feature Scaling

Before clustering, all features are standardized using Z-score normalization to ensure comparable contribution

across dimensions. For feature j , standardization is defined as:

$$z_j = \frac{(x_j - \mu_j)}{\sigma_j}$$

where μ_j and σ_j are the mean and standard deviation of feature j computed from the training dataset. During deployment, the same μ and σ values are reused to guarantee consistency between offline training and live inference.

D. K-Means Clustering

K-Means partitions N samples into k clusters by minimizing the within-cluster sum of squared distances. The objective function is:

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

where C_i denotes the set of samples in cluster i and μ_i is the centroid of that cluster [2]. The algorithm iteratively updates centroids and assignments until convergence. Multiple random initializations are performed, and the solution with minimum inertia is retained to reduce sensitivity to initialization.

4.1. SELECTING THE NUMBER OF CLUSTERS (K)

Selecting k is a critical design decision. WARTEQ adopts a two-stage approach combining visual inspection and quantitative validation.

A. Elbow Method

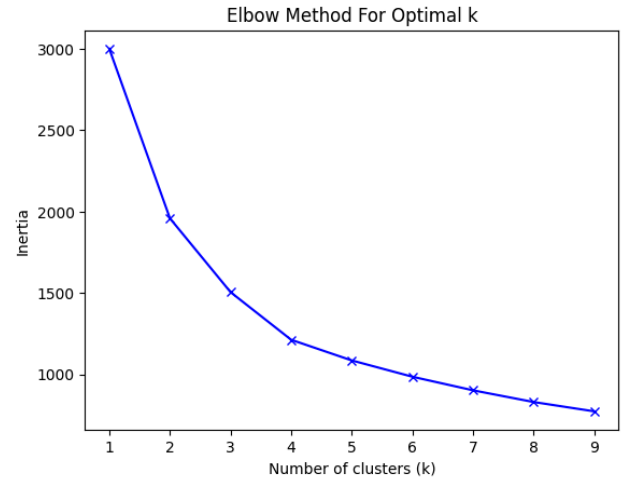


Fig.2 Elbow Method graph for Optimal K

The Elbow Method evaluates inertia as a function of k . As shown in **Fig. 2**, inertia decreases sharply from $k = 1$ to $k = 3$, after which the improvement diminishes. This “elbow” indicates that three clusters capture the dominant structure without over-fragmentation.

B. Metric Cross-Check

K	Silhouette Score	Davies-Bouldin	Calinski-Harabasz
2	0.3169	1.2386	264.7
3	0.3089	1.1310	264.7
4	0.3045	1.1756	243.6
5	0.2788	1.2231	218.1

Table.1 Accuracy metric comparison table

Internal validity metrics are computed for $k = 2$ to 5, as summarized in **Table 1**. Although $k = 2$ achieves a marginally higher Silhouette score, $k = 3$ yields the lowest Davies–Bouldin Index and aligns naturally with three interpretable regimes (low, medium, high). The Calinski–Harabasz Index further supports this choice.

4.2. CLUSTERING EVALUATION AND INTERPRETATION

Because clustering is unsupervised, traditional classification accuracy is undefined. Instead, internal validation metrics are used.

A. Silhouette Coefficient

For sample i , the Silhouette coefficient is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the minimum mean distance to samples in another cluster [4]. Values closer to 1 indicate better separation. Observed values (~ 0.28 – 0.32) indicate moderate but meaningful separation for real-world environmental data.

B. Davies–Bouldin Index

The Davies–Bouldin Index measures cluster similarity based on intra-cluster dispersion and inter-cluster distance. Lower values indicate better clustering [5]. As shown in **Table I**, the minimum DBI occurs at $k = 3$.

C. Calinski–Harabasz Index

The Calinski–Harabasz Index compares between-cluster variance to within-cluster variance [6]. Higher values indicate well-separated, compact clusters. The observed decrease beyond $k = 3$ further supports the selected configuration.

4.3. VISUALIZATION VIA PCA AND 2D PLOTS

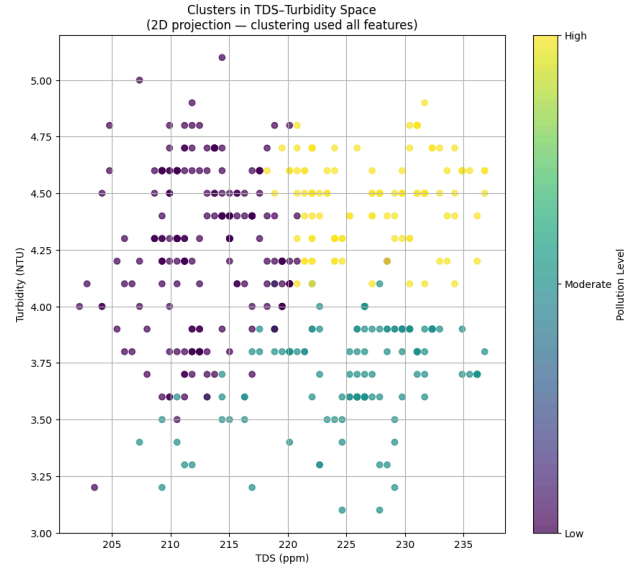


Fig.3 PCA plot for clusters visualization

A two-dimensional scatter plot of TDS versus turbidity **Fig. 3** highlights clear separation between clusters, reflecting differences in dissolved and suspended solids. To further examine structure in higher dimensions, Principal Component Analysis (PCA) is applied.

Given standardized data matrix X , PCA computes eigenvectors of the covariance matrix and projects data as:

$$Z = X \cdot W$$

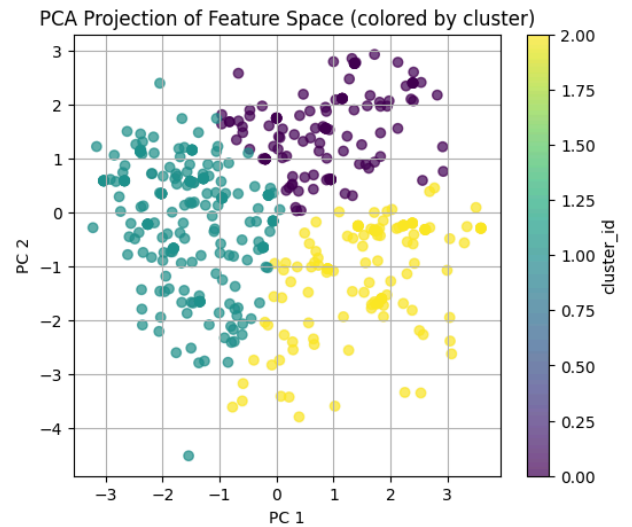


Fig.4 PCA plot projection for Feature Space

where columns of W are principal directions [8]. **Fig. 4** shows the projection onto the first two principal components, confirming that clusters occupy distinct regions in the dominant variance subspace.

4.4. DEPLOYMENT IN WEB DASHBOARD

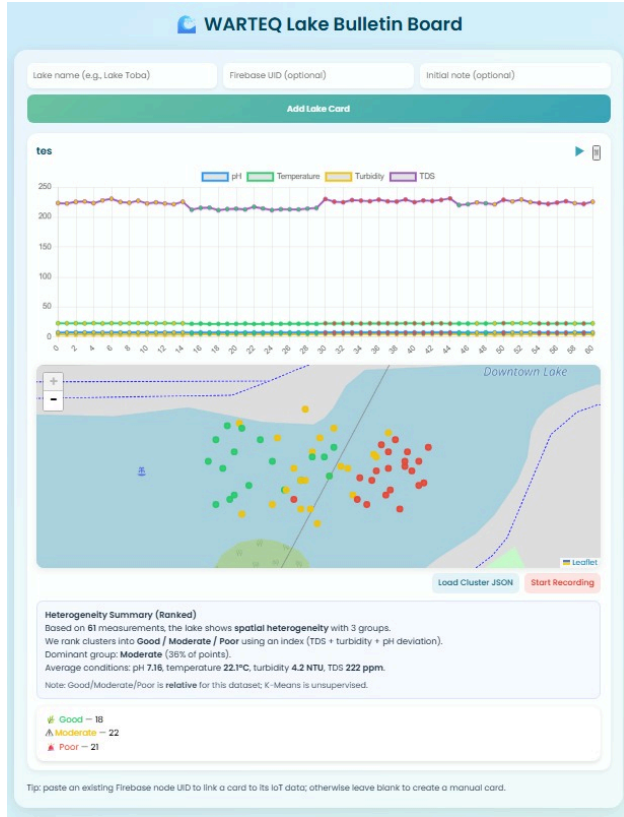


Fig.5 Website output

To enable real-time inference without server-side machine learning dependencies, scaler parameters (μ , σ) and K-Means centroids are embedded directly into the web application. Incoming measurements are transformed using the same preprocessing pipeline and assigned to the nearest centroid using Euclidean distance. For user interpretability, clusters are optionally ranked as Good, Moderate, or Poor using a relative index based on standardized TDS, turbidity, and pH deviation. **Fig. 5** shows the deployed dashboard interface and spatial clustering output.

5. CONCLUSION

This paper presented the machine learning pipeline of WARTEQ, an RC-boat-based system for spatial clustering of lake water-quality measurements. By leveraging unsupervised learning, WARTEQ identifies natural groupings in sensor data without requiring labeled ground truth, making it well suited for exploratory environmental monitoring and rapid field deployment.

The proposed approach integrates data cleaning, feature engineering, Z-score normalization, and K-Means clustering, with the number of clusters selected through a combination of elbow analysis and internal validation metrics. The use of both raw and logarithmically transformed features enables the model to capture absolute pollution magnitude while remaining robust to skewed distributions. Experimental results demonstrate that three clusters provide a meaningful and interpretable separation of water-quality regimes, supported by consistent trends across Silhouette, Davies–Bouldin, and Calinski–Harabasz indices.

A key contribution of WARTEQ lies in its deployment strategy. By exporting scaler parameters and cluster centroids to a browser-based dashboard, the system performs real-time inference without server-side machine learning dependencies. This design enables interactive visualization of spatial heterogeneity, allowing users to identify relative “Good,” “Moderate,” and “Poor” water-quality zones within a given survey. Importantly, these labels represent relative rankings rather than regulatory classifications, emphasizing the system’s role as a decision-support and exploratory analysis tool.

Overall, WARTEQ demonstrates that lightweight, interpretable machine learning methods can effectively support spatial water-quality monitoring when combined with mobile sensing platforms. The proposed pipeline provides a practical foundation for rapid assessment of lake conditions and can be readily extended with additional sensors, validation strategies, and advanced clustering techniques to further enhance environmental insight and decision-making.

REFERENCES

- [1] World Health Organization, *Guidelines for Drinking-Water Quality*, 4th ed. Geneva, Switzerland: WHO, 2017.
- [2] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. Math. Statist. Prob.*, 1967, pp. 281–297.
- [3] C. Wongoutong, "The impact of neglecting feature scaling in K-means clustering," *PLoS ONE*, vol. 19, no. 12, Art. e0310839, 2024.
- [4] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *J. Comput. Appl. Math.*, vol. 20, pp. 53–65, 1987.
- [5] D. L. Davies and D. W. Bouldin, "A cluster separation measure," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. PAMI-1, no. 2, pp. 224–227, 1979.
- [6] T. Calinski and J. Harabasz, "A dendrite method for cluster analysis," *Commun. Stat.*, vol. 3, no. 1, pp. 1–27, 1974.
- [7] Apera Instruments, "What is Conductivity, Resistivity, TDS, Salinity, and Their Relationship?" [Online]. Available: <https://aperainst.com/blogs/aperablog/what-is-conductivity-resistivity-tds-salinity-and-their-relationship>. Accessed: Jan. 5, 2026.
- [8] I. T. Jolliffe and J. Cadima, "Principal component analysis: A review and recent developments," *Philos. Trans. R. Soc. A*, vol. 374, no. 2065, 2016.
- [9] Fondriest Environmental, "Turbidity, Total Suspended Solids & Water Clarity," 2014. [Online]. Available: <https://www.fondriest.com/environmental-measurements/parameters/water-quality/turbidity-total-suspended-solids-water-clarity/>.
- [10] C. Meng, "Water Quality Testing Dataset," GitHub repository. [Online]. Available: <https://github.com/ChanMeng666/water-quality-testing-data-analysis>. Accessed: Jan. 5, 2026.